

INTEGRATED TRL – MRL – DECISION STAGE GATES

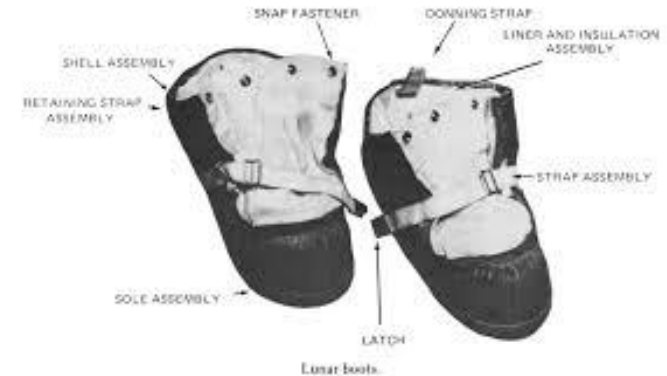
FINAL PRESENTATION



Space Boots

Boots – Parts

- **Inner Boot**
 - Provides support, flexibility, and injury protection
 - Enables natural foot motion and integrates with cooling
 - Prevents pressure bladder wrinkles and fits snugly into the outer boot
- **Outer Boot**
 - Functions as a pressure vessel or restraint layer
 - Supports structural integrity and thermal regulation
 - Interfaces with sealing system and enhances mobility and protection
- **Overboot**
 - Shields from lunar regolith, extreme temperatures, and abrasion
 - Enhances traction on the lunar surface
- **Cooling Integration**
 - Interfaces with Liquid Cooling Garment (LCG) for thermal control
 - Options include active (electric/liquid) or passive insulation systems
- **Sealing System**
 - Ensures airtight interface between the boot and the lower leg suit



Boots – TRL

TRL 5: Component/Material Validation

- Vacuum, flexibility, dust, abrasion, impact, water resistance

TRL 6: System Integration Testing

- Microgravity, thermal cycles, suit integration, EMI, compression testing

TRL 7: Operational Simulation

- Analog missions, EVA simulations, cold soak flexibility

TRL 8: Final Qualification

- Pre-commercial tests, extended wear, flight certification

TRL 9: Operational Use

- Space mission use, post-mission validation, commercial release
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Boots – MRL

MRL 4: Prototype Manufacturing Validation

- Material cutting, stitching, zipper installation, thermal liner assembly

MRL 5: Assembly and Fit Validation

- Adhesive bonding, dust sealing, sole mold fit, insulation placement

MRL 6: Durability and Endurance Testing

- Thermal cycling, wear cycling, tread wear, sealing under flex

MRL 7: Field and Operational Testing

- EVA field use, pressure sealing, long-term wear, don/doff testing

MRL 8: Pre-Flight and Production Readiness

- Flight qualification, batch production, extended thermal testing, user acceptance

MRL 9: In-Mission and Post-Mission Validation

- Space mission use, QA audit, client feedback, post-flight recovery

MRL 10: Mass Production and Scalability

- Commercial production review, scalability check, automation integration
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Boots – Recommendation / Improvements

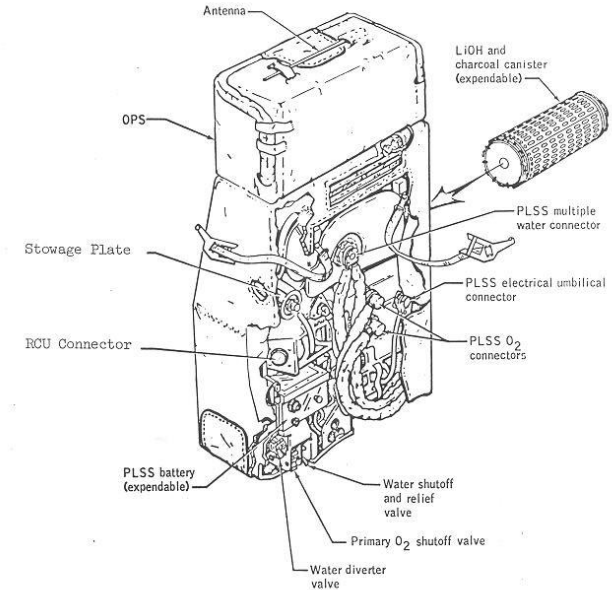
- **Enhanced Dust Mitigation**
 - Add electrostatic dust repellent layers to reduce lunar regolith buildup
- **Thermal Insulation Upgrade**
 - Integrate phase-change materials for passive temperature control
- **Improved Sealing Mechanism**
 - Explore magnetic or shape-memory alloy seals for faster, more secure attachment
- **Tread Optimization**
 - Test advanced outsole geometries for better grip on varied lunar surfaces
- **Material Durability**
 - Evaluate abrasion-resistant fabrics or longer wear life



Life Support

Life Support – Parts

- **Oxygen Supply**
 - High-pressure tank with flow control for regulated delivery
- **CO₂ Removal**
 - LiOH or sorbent filters; automated CO₂ level sensors
- **Cooling System**
 - LCG + Sublimator + ATCL for heat rejection and backup
- **Ventilation**
 - Airflow fans and dehumidifiers for moisture control
- **Remote Control Unit**
 - User interface with system status and alerts
- **Structure & Power**
 - Lightweight backpack frame; compact battery
- **Emergency Systems**
 - Backup oxygen purge and manual overrides



Life Support – TRL

TRL 5: Component Validation

- Oxygen tank pressure, CO₂ scrubber efficiency, battery endurance, vibration, EMI, ESD, sensor calibration, structural integrity

TRL 6: Subsystem Validation

- Emergency backup activation, thermal regulation, humidity control, hydrostatic pressure, external load

TRL 7: System Prototype Demonstration

- Long-duration performance, communication, contamination resistance, integrated EVA sim, field humidity control, emergency oxygen activation

TRL 8: Integrated System Evaluation

- Sensor network stability, suit seal inspection, battery power cycle, post-sim inspection, electrical system test

TRL 9: Mission-Ready Operations

- Leak detection, analog mission evaluation, full EVA simulation, suit performance, post-mission evaluation
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Life Support – MRL

MRL 4: Prototype Manufacturing Validation

- CO₂ scrubber fabrication, O₂ tank machining, humidity module build, regulator valve machining, connector welding, tubing integration, power connector assembly

MRL 5: Assembly and Fit Validation

- Integrated subsystem assembly, modular connect testing, thermal uniformity, vibration testing, pressure decay testing, EMI pre-scan

MRL 6: Durability and Endurance Testing

- Environmental stress testing, thermal vacuum exposure, rapid depressurization, humidity cycling, oxygen exposure, pressure cycling, emergency shutdown testing

MRL 7: Field and Operational Testing

- Limited-rate production build, kitting efficiency, first article inspection, pilot lot reliability testing

MRL 8: Pre-Flight and Production Readiness

- Full-rate production setup, long-term validation, final stress testing, configuration audit, cleanliness validation

MRL 9: In-Mission and Post-Mission Validation

- Flight certification, human factors testing, redundancy activation, mission simulation

MRL 10: Mass Production and Scalability

- Production sustainment audit, fleet monitoring, continuous improvement loop, long-term reliability tracking

Life Support – Recommendation / Improvements

- **CO₂ Scrubber Efficiency**
 - Implement reusable or regenerable sorbents to extend mission duration
- **Battery Life Extension**
 - Use next-gen lithium-silicon batteries for higher energy density
- **Humidity Control Enhancements**
 - Improve dehumidifier capacity and airflow distribution
- **System Interface Upgrade**
 - Develop a more intuitive, glove-friendly Remote Control Unit
- **Redundancy Expansion**
 - Add secondary fans/pumps to increase fault tolerance during EVA

Questions ?